UNIT – 2	9 Hours
Analysis of AC circuits	
Instantaneous value, RMS value(Only for sine wave), Average value,(Only for half sine wave) Form factor, Peak factor. **Single-phase AC Circuits:** Active power, power factor, phase, Analysis of R, L,C(Discussion only on active power) Analysis of RL, RC.((Discussion only on active power) Analysis of RLC Circuits with phasor diagram.	

2.1.1 Analysis of AC Circuits:

Alternating current (AC) : is an electric current that periodically reverses direction and changes its magnitude continuously with time.

Some of the examples of AC waveform are shown in Figure 2.1

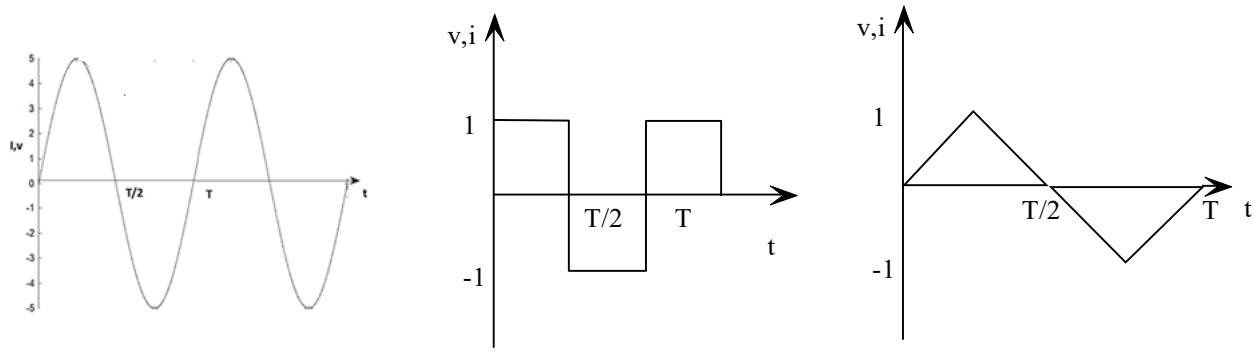


Fig 2.1 Examples of AC waveforms

2.1.2 Terminologies used in AC sine wave representation: ωt /(θ)

Consider the sine wave shown in Figure 2.2

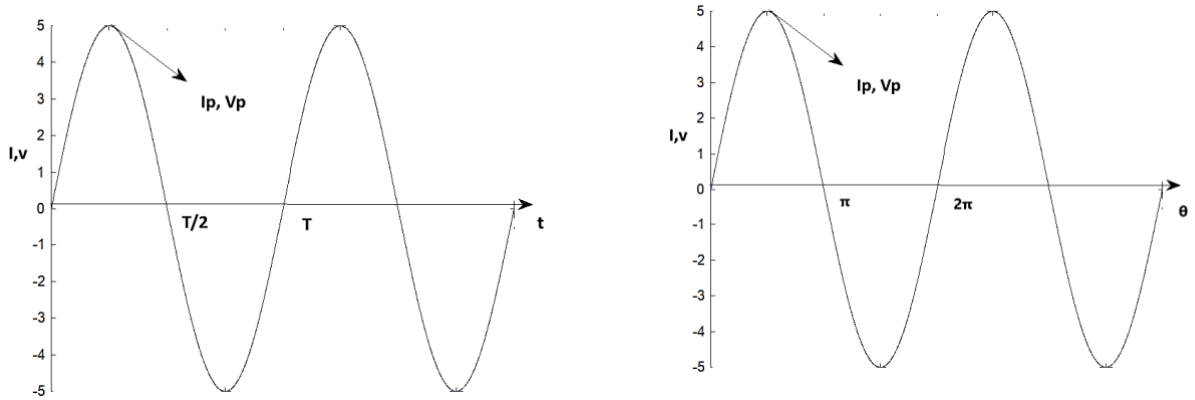


Fig 2.2 Sinusoidal voltage or current waveform

(i) **Cycle:** When an Alternating quantity goes through complete set of positive and negative values or goes through 360 electrical degrees.

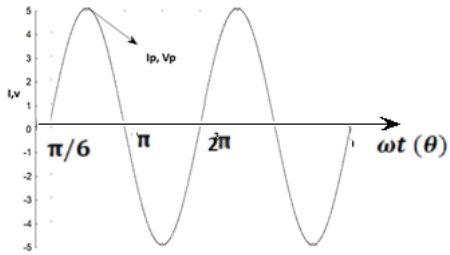
(ii) **Time Period (T) :** Time taken to complete one cycle by AC wave. It is represented by T (sec)

(iii) **Frequency (f) :** Number of cycles made per second. It is the reciprocal of Time period ($\frac{1}{T}$) and its unit is Hz(Hertz)

(iv) **Amplitude:** Maximum value attained by an alternating quantity in one cycle and also called Peak Value / Max. Value.

(v) **Phase:**

If the sine wave is shifted by π/6 from the zero point of reference as shown in Figure 2.3, then phase is said to be π/6 radians



2.1.3 Instantaneous value: The magnitude of a waveform at any instant of time with reference to zero. Instantaneous values are denoted by lower case symbols such as e, v and i etc.

Consider the current waveform shown in Fig 2.3

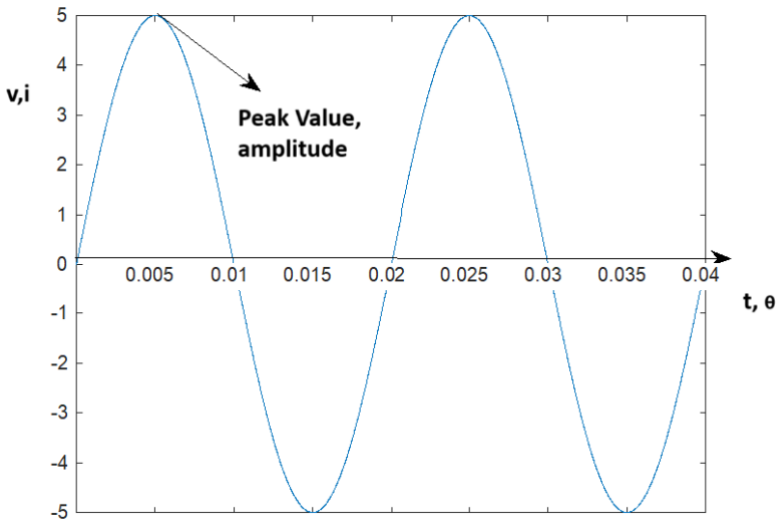


Fig 2.3 Sinusoidal voltage or current wave

The waveform is defined as $i = 5 * \sin(\omega t) = 5 * \sin(314t)$or $v = 5 * \sin(314t)$

where $\omega = 2 * \pi * f = 2 * \pi * \frac{1}{T}$ and its unit is rad/sec

5 is the amplitude of the wave.

Instantaneous current at

t=0, i=0

t=0.005 S, i=5A

t=0.01 S, i=0A

t=-015 S, i=-5A

2.1.4 RMS value:

"RMS" stands for "Root-Mean-Squared", also called the effective or heating value of alternating current.

If $i = I_m \sin\theta$ is passed through a resistor having resistance R ohms, the heating effect of 'i' is i^2R.

Thus, the RMS value of an alternating current is defined as that steady(D.C) current which when flowing through a given resistor for a given time produces the same amount of heat as produced by the alternating current when owing through the same resistor for the same period of time.

In other words, the R.M.S value is defined as the square root of means of squares of instantaneous values.

Consider the current waveform shown in Figure 2.4

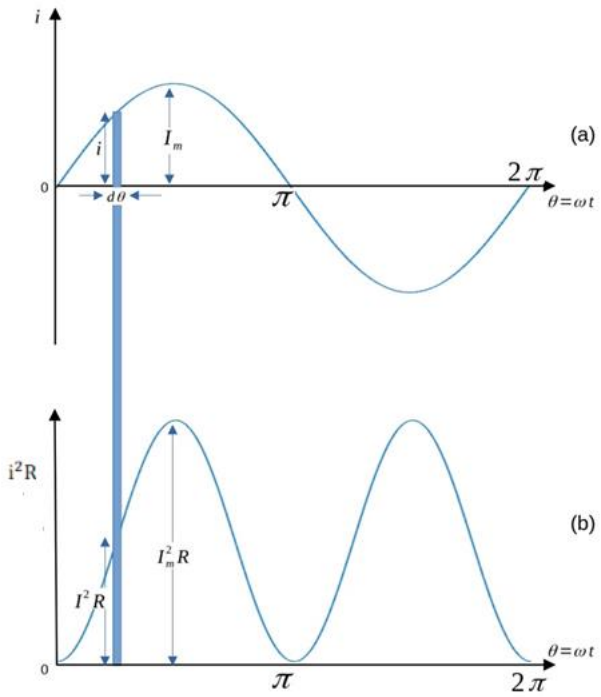


Fig 2.4 i and i^2R wave

If the current is passed through a resistor having resistance R ohms, instantaneous heating effect is given by bi^2R watts.

Average heating effect over half-cycle is

$$= \frac{\text{Area enclosed by } i^2R \text{ over half cycle}}{\text{Length of base}}$$

Consider a small elementary strip of length dθ in i^2R wave as shown in Figure 3.4 (b). During interval dθ radians, heat generated is i^2R dθwatt radians and is represented by the area of the shaded strip. Henceheat generated during the first half-cycle is the area enclosed by the i^2R curve and is given by

$$\int_0^{\pi} i^2R\, d\theta = I_m^2 R \int_0^{\pi} \sin^2\theta\, d\theta == I_m^2 R \int_0^{\pi} \left(\frac{1-\cos 2\theta}{2}\right) d\theta$$

$$= I_m^2 R \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_0^{\pi} = \pi R \frac{{I_m}^2}{2}$$

$$=$$

Then Average Heating effect is $\frac{\pi R \frac{{I_m}^2}{2}}{\pi} = R\frac{I_m^2}{2}$

If 'I' is the value of direct current through the same resistance to produce the same heating effect as that of ac

$$I^2R = R\frac{I_m^2}{2}$$

Or $$I = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

Note: The above derivation is valid for Voltage also. $V = \frac{V_m}{\sqrt{2}}$

2.1.5 Average Value:

This is defined on the basis of amount of charge transferred, which is given by q = I t.

The average value of an alternating current is equal to that (D.C) steady current, which transfers the same amount of charge as transferred by the alternating current across the same circuit and in the same time.

Or, alternatively, average value of current is

$$\frac{\text{Area enclosed over a cycle}}{\text{Length of base over a cycle}}$$

Average value of a sinusoidal wave: The average value of a sinusoidal wave is always zero.

Average value of a half rectified sinusoidal wave:

Consider the rectified sinusoidal wave with an elemental rectangular strip of length dθ as shown in Figure 2.5

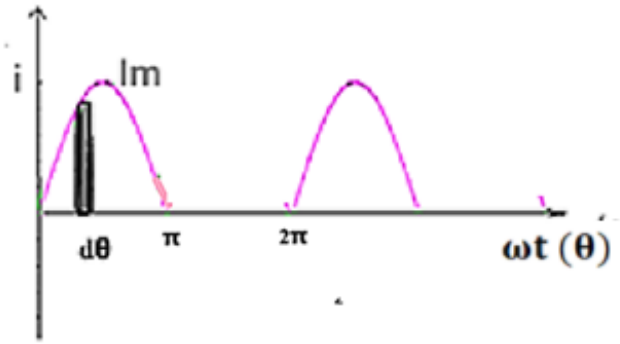


Fig 2.5 Half Rectified sine wave

For a very small interval dθ radians, the area of the shaded strip is i.dθ ampere radians. Therefore, total area enclosed by the current wave over half-cycle is

$$\int_0^{\pi} i\, d\theta = \int_0^{\pi} I_m \sin\theta\, d\theta = I_m[-\cos\theta]_0^{\pi} = 2I_m \text{ ampere radians}$$

Thus, average value of current is

$$\frac{\text{Area enclosed over a cycle}}{\text{Length of base over cycle}} = \frac{2I_m}{2\pi} = 0.318\, I_m$$

Note: Similarly, the average voltage is given by$V_{av} = 0.318\, V_m$

2.1.6 Form Factor (K_f):

The ratio of RMS value to average value of an alternating quantity is called formfactor. For fully rectified sine wave current

$$K_f = \frac{RMS\ value}{Average\ value} = \frac{\frac{I_m}{\sqrt{2}}}{\frac{2I_m}{\pi}} = 1.11$$

2.1.7 Peak Factor K_p :

The ratio of maximum value to the RMS value of an alternating quantity is called peak factor. For fully rectified sine wave current

$$K_p = \frac{Max\ value}{RMS\ value} = \frac{I_m}{I} = \frac{I_m}{0.707\, I_m} = 1.414$$

2.2 Power and Power factor:

Power in AC circuits is defined as the amount of energy delivered per unit of time by the circuit. A circuit element dissipates or produces power according to P=IV, where I is the current through the element and V is the voltage across it. Since the current and the voltage both depend on time in an ac circuit, the instantaneous power p(t)=i(t)v(t) is also time dependent.

2.2.1 Active power (Real Power): This is the power that performs useful work, such as running machinery, lighting, or heating equipment. It is given by P=VI cos(∅) and its unit is Watt (W), KW or MW.

2.2.2 Power factor: Cosine of the angle between the voltage and current in the circuit is known as the power factor of the alternating current. Power factor (PF) is a dimensionless quantity, ranging from 0 to 1, that measures how effectively an electrical system uses the power it receives.
It is given by cos(∅) where ∅ is the angle by which current lags the voltage or advances the voltage (popularly known as phase angle of the circuit).

2.2.3 Phase: It is the angle by which current lags the voltage or advances the voltage in a circuit.

2.3 Single-phase AC circuits

2.3.1 Analysis of a resistive circuit:

Consider a circuit having resistance R ohms connected across the terminals of an a.c. supply, as in Figure 2.6. In the circuit shown R is a pure resistance to which an alternating voltage, $v = V_m \ sin\omega t$ is applied, due to which an alternating current i flows through it.

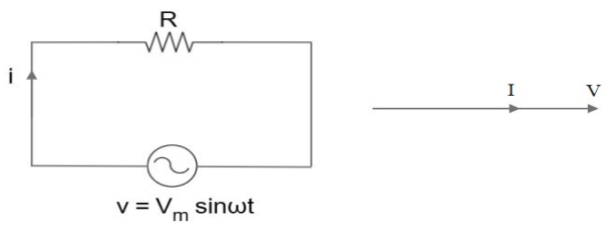


Fig 2.6 AC sinusoidal voltage applied to Resistance R

By Ohms law,

$$i = \frac{v}{R} = \frac{V_m \sin(\omega t)}{R} = I_m \sin(\omega t) \ \text{ where, } I_m = \frac{V_m}{R}$$

By observing the equations for voltage, v=$\mathbf{V_m}$sinωt and i= $I_m$sinωt,it can be concluded that the current is in phase with the voltage. Vectorially the RMS values of voltage and current are represented as shown in Figure 3.9

The instantaneous power consumed by the resistance is given by,

$$p = v \times i = V_m \sin\omega t I_m \sin\omega t \ = V_m I_m \sin^2\omega t$$

$$p = V_m I_m \left(\frac{1 - \cos 2\omega t}{2}\right) = \frac{1}{2} V_m I_m - \frac{1}{2} V_m I_m \cos 2\omega t$$

Equation consists of two parts. The second part $\frac{1}{2} V_m I_m cos2\omega t$is a periodicallyvarying quantity whose frequency is two times the frequency of the applied voltage and its average value over a period of time is zero. Power is a scalar quantity and hence only its average value has to be taken into account. Hence the power consumed by the resistance is only due to the first part$\frac{1}{2} V_m I_m$.

∴ Average Power,

$$P = \frac{1}{2} V_m I_m = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} = VI$$

2.3.2 Analysis of a inductive circuit:

Consider a circuit with a coil of pure inductance L Henry connected across the terminals of an AC supply as shown in Figure 2.7

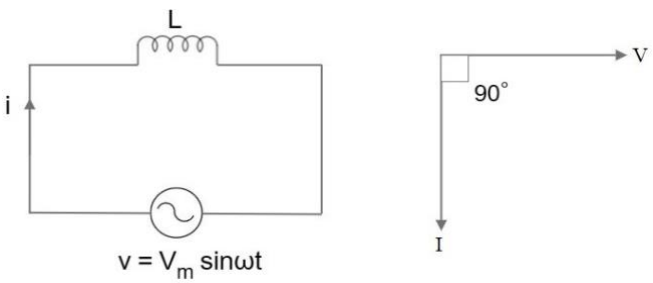


Fig 2.7 Pure inductive circuit

Due to the applied alternating voltage $v=V_m\sin(\omega t)$, an alternating current i flows through it. This alternating current produces an alternating flux, which links the coil and hence an e.m.f e is induced in it, which opposes the applied voltage and is given by,

$$e = -L \frac{di}{dt} = -v$$

$$\therefore v = L \frac{di}{dt}$$

$$di = \frac{v}{L} dt = \frac{1}{L} V_m \sin(\omega t)\, dt$$

Integrating the above equation, we get

$$i = \frac{V_m}{L} \int \sin(\omega t)\, dt$$

$$= \frac{V_m}{\omega L} (-\cos(\omega t))$$

$$= \frac{V_m}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right)$$

$$i = I_m \sin\left(\omega t - \frac{\pi}{2}\right)$$

where $$I_m = \frac{V_m}{\omega L} = \frac{V_m}{X_L}$$

$$X_L = \omega L = 2\pi f L = reactive\ inductance\ in\ Ohms$$

By observing the equations for voltage and current, we find that the current lags the voltage by an angle$\pi/2$.VectorialrepresentationoftheRMSvaluesofvoltageand current is as shown in Figure 3.12 (b)

The instantaneous power is given by,

$$p = v.i = V_m \sin\omega t . I_m \sin\left(\omega t - \frac{\pi}{2}\right)$$

$$p = V_m I_m \sin\omega t(-\cos\omega t) = -\frac{1}{2} V_m I_m \sin 2\omega t$$

The equation for p consists of a quantity which is periodically varying and having a frequency two times the frequency of the applied voltage and whose average value is zero. The power consumed by a pure inductor is zero.

2.3.3 Analysis of a capacitive circuit:

Consider a capacitor of pure capacitance C, across which an alternating voltage v=$V_m sin\omega t$ is applied, due to which an alternating current I flows, charging the plates of the capacitor with a charge of q coulombs as shown in Figure 2.8

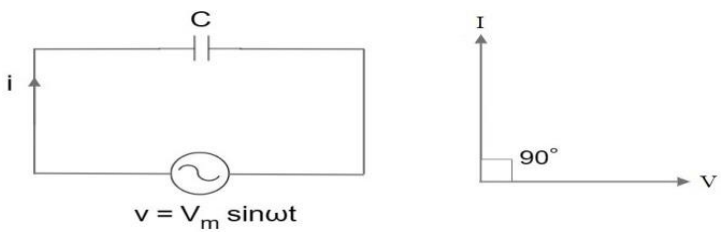


Fig 2.8 Pure capacitive circuit

The current 'i' flowing through the circuit is given by

$$i = \frac{dq}{dt} = \frac{d(Cv)}{dt} = C\frac{dv}{dt} = C\frac{d}{dt}(V_m \sin(\omega t))$$

$$= \omega C V_m \cos\omega t = \frac{V_m}{1/\omega C}\sin\left(wt + \frac{\pi}{2}\right)$$

$$= \frac{V_m}{X_c}\sin\left(\omega t + \frac{\pi}{2}\right)$$

$$i = I_m \sin(\omega t + \pi/2)$$

Where $I_m = \frac{V_m}{X_c}$ $and\ X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$ = Capacitive reactance in Ohms.

By observing the equations for voltage and current, we find that the current leads the voltage by an angle $\pi/2$.

The equation for p consists of a quantity which is periodically varying and having a frequency two times the frequency of the applied voltage and whose average value is zero. The power consumed by a pure capacitor is zero.

Exercise 1: An AC Circuit consist of a pure resistance of 20 Ω and is connected across an sinusoidal AC supply of 230 V, 50Hz.Calculate current, power consumed, equation for voltage and current.

Solution:

I=V/R=230/20=11.5A

P=VI=230×11.5=2645W

$$V_m = \sqrt{2}V = 325.27V$$

$$I_m = \sqrt{2}I = 16.26A$$

$\omega = 2\pi f = 314Hz$

$v = V_m \sin\omega t = 325.27\sin(314t)$ V

$i = I_m \sin\omega t = 16.26\sin(314t)$ A

Exercise 2: An inductive circuit allows a current of 20 A to flow from 230V, 50Hz supply. Calculate inductive reactance, inductance of the coil, power absorbed, equations for voltage and current.

Solution:

Inductive reactance, $X_L = V/I = 230/20 = 11.5\Omega$

Coil inductance ,$L = X_L/2\pi f = 11.5/2\pi*50 = 0.036H$

Power absorbed = 0

Peak value of the voltage, $V_m = \sqrt{2}V = \sqrt{2} * 230 = 325.27V$

Peak value of the current,$I_m = \sqrt{2}I = \sqrt{2} * 20 = 28.28A$

$\omega = 2\pi f = 314Hz$

Equation of the voltage :$v = V_m \sin(\omega t) = 325.27\sin 314t$ V

Equation of the current : $i = I_m(\sin\omega t - \frac{\pi}{2}) = 28.28\sin(314t - \frac{\pi}{2})A$

Exercise 3: A 400μF capacitor is connected across a 230V, 50Hz supply. Calculate capacitive reactance, rms current, equations for the voltage and current.

Capacitive reactance=$X_C = 1/(2\pi fC) = 1/(2\pi * 50 * 400 * 10^{-6}) = 7.95\Omega$

RMS current, $I = V/X_C = 230/7.95 = 28.93A$

Peak value of the voltage, $V_m = \sqrt{2}V = \sqrt{2} * 230 = 325.27V$

Peak value of the current, $I_m = \sqrt{2}I = \sqrt{2} * 28.93 = 40.91A$

$\omega = 2\pi f = 314Hz$

Equation of th voltage: $v = V_m \sin(\omega t) = 325.27 \sin 314t$

Equation of the current: $i = I_m \left(\sin\left(\omega t + \frac{\pi}{2}\right)\right) = 40.91 \sin(314t + \pi/2)$ A

2.3.4 Analysis of a R-L Series circuit:

Consider an R-L series circuit as shown in Figure 2.9. An alternating voltage of $v = V_m \sin\omega t$ is applied, due to which an alternating current i flows through the circuit. The vector diagram considering I as reference vector is as shown inFigure.2.9.

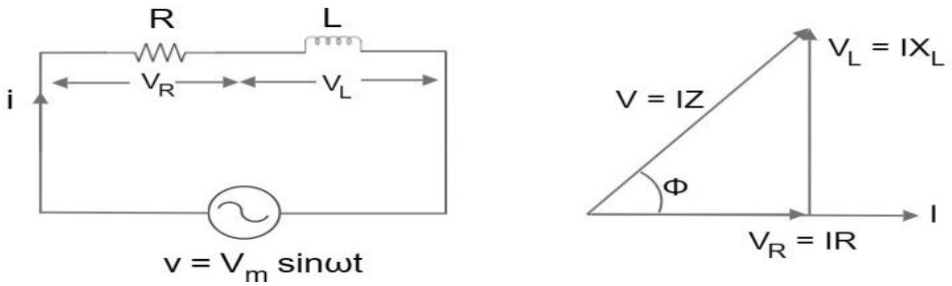


Fig 2.9 : RL series circuit

The vector diagram consists of three voltages,

$V_R = I R$ which is in phase with the current,

$V_L = I X_L$ which leads the current by 90.

The vector sum of these two voltages is applied voltage V=I Z. here, Z is the impedance of the circuit in ohms

The impedance of an a.c circuit may be defined as the opposition offered for the flow of alternating current in the circuit. It is the phasor sum of resistance and reactance of the circuit. The impedance triangle for an R-L series circuit is shown in Figure 2.10

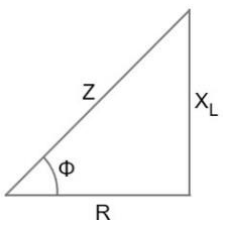


Fig 2.10 :Impedance triangle

$V = V_R + V_L$ (phasor sum)

$$V = \sqrt{V_R^2 + V_L^2} = \sqrt{I^2R^2 + I^2{X_L}^2}$$

$= I\sqrt{R^2 + X_L^2} = IZ$

where, $Z=\sqrt{R^2 + X_L^2}$ and $\varphi = \tan^{-1}\left(\frac{X_L}{R}\right) = \cos^{-1}\left(\frac{R}{Z}\right)$

From the phasor diagram shown in Figure 2.9 ,we observe that the current lags the voltage by an angle $\emptyset$.

If $v = V_m \sin\omega t$

$i=I_m(\sin(\omega t)-\emptyset)$

The instantaneous power is given by,

$p=v.i= V_m \sin\omega t I_m(\sin\omega t - \emptyset)$

$=V_m I_m \frac{1}{2}[\cos\varphi - \cos(2\omega t - \emptyset)]$

$p=\frac{1}{2}V_m I_m \cos\varphi - \frac{1}{2}V_m I_m \cos(2\omega t - \emptyset)$

The second term in the equation is a periodically varying quantity, whose frequency is two times the frequency of the applied voltage and its average value is zero. As power is always an average value, only the first term represents the power consumed.

$\therefore P=\frac{1}{2}V_m I_m \cos\emptyset = \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}} \cos\emptyset=VI \cos\emptyset$

2.3.5 Analysis of a R-C Series circuit

Consider an R-C series circuit as shown in Figure 2.11. An alternating voltage of v =V_m sinωt is applied, due to which an alternating current i flows through the circuit. The vector diagram considering I as reference vector is as shown inFigure.2.11.

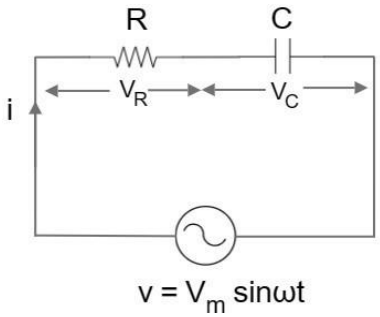
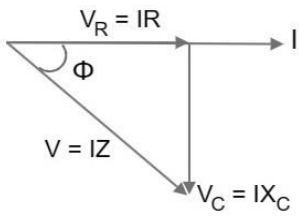


Fig 2.11 : RC series circuit



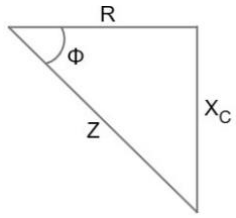


Fig 2.12 :Impedance Diagram

V_R= I R which is in phase with the current, V_C=I X_C which lags the current by 90°.The vector sum of these two voltages is applied voltage V=IZ. Here, Z is the impedance of the circuit in ohms. The impedance triangle for an R-C series circuit is shown in Figure 2.12.

$V=V_R+V_C$ (phasor sum)

$V = \sqrt{V_R^2 + V_c^2} = \sqrt{I^2R^2 + I^2X_C^2}$

$= I\sqrt{R^2 + X_C^2} = IZ$

Where, $Z = \sqrt{R^2 + X_C^2}$ and $\varphi = tan^{-1}\left(\frac{X_c}{R}\right) = cos^{-1}\left(\frac{R}{Z}\right)$

The current leads the voltage by an angle.

If $v=V_m\sin\omega t$, Then, $i= I_m\sin(\omega t+ \varphi)$

The current in the circuitis given by, I=V/Z

The impedance triangle for an R-C series circuit is shown in Figure 3.20. The instantaneous power is given by,

$p=v.i= V_m\sin(\omega t).I_m(\sin(\omega t + \emptyset)$

$=V_mI_m\frac{1}{2}[\cos(-\emptyset) - \cos(2\omega t + \emptyset)]$

$$p = \frac{1}{2}V_mI_m\cos(\emptyset) - \frac{1}{2}V_mI_m\cos(2\omega t + \emptyset)$$

The second term in the equation is a periodically varying quantity, whose frequency is two times the frequency of the applied voltage and its average value is zero. As power is always an average value, only the first term represents the power consumed.

$\therefore P=\frac{1}{2}V_mI_m\cos\emptyset = \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\cos\emptyset = VI\cos\emptyset$

2.3.6 Analysis of a R-L-C Series circuit

Consider an R-L-C series circuit as shown in Figure.2.13.to which an alternating voltage of $v=V_m\sin\omega t$ is applied, due to which an alternating current I flows through the circuit. Three cases of the circuit can be discussed here.

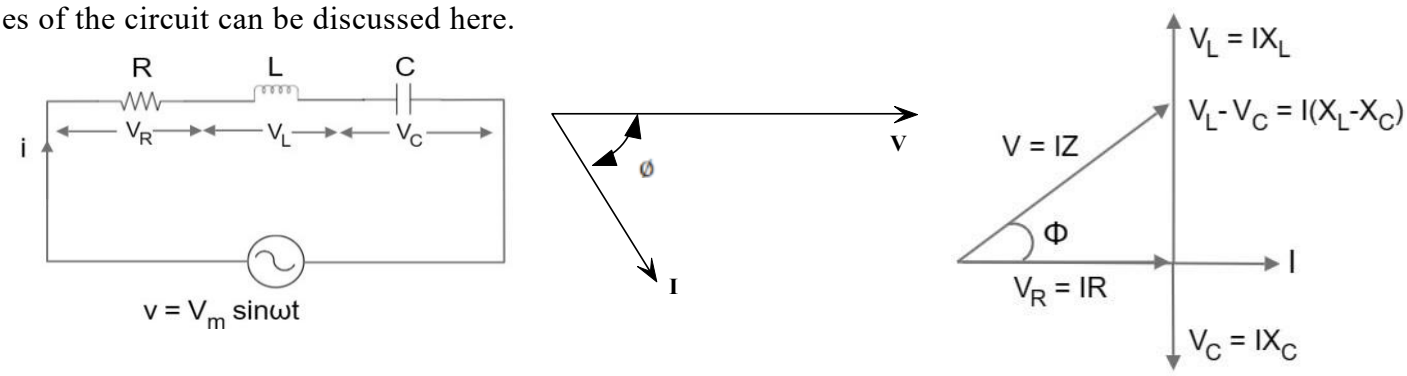


Fig2.13: a)RLC series circuit b)Phasor Diagram for $X_L>X_C$ c) Impedance triangle

Case1 : When $X_L > X_C$

When the inductive reactance is more than the capacitive reactance, the vector diagram of the circuit as shown in Figure 2.13b. from the vector diagram we observe that the current lags the voltage by an angle Ø.The impedance triangle is shown in Figure 2.13c.

$$I = \frac{V}{Z}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

The circuit behaves similar to an R-L circuit. If $v = V_m \sin(\omega t)$, then $i = I_m \sin(\omega t - \emptyset)$

Hence it can be proved that the power consumed is given by $P = VI\cos\emptyset$.

Case 2: When $X_L < X_C$

When the inductive reactance is less than the capacitive reactance, the vector diagram of the circuit as shown in Figure2.14a. From the vector diagram, we observe that the current leads the voltage by an angle Ø.

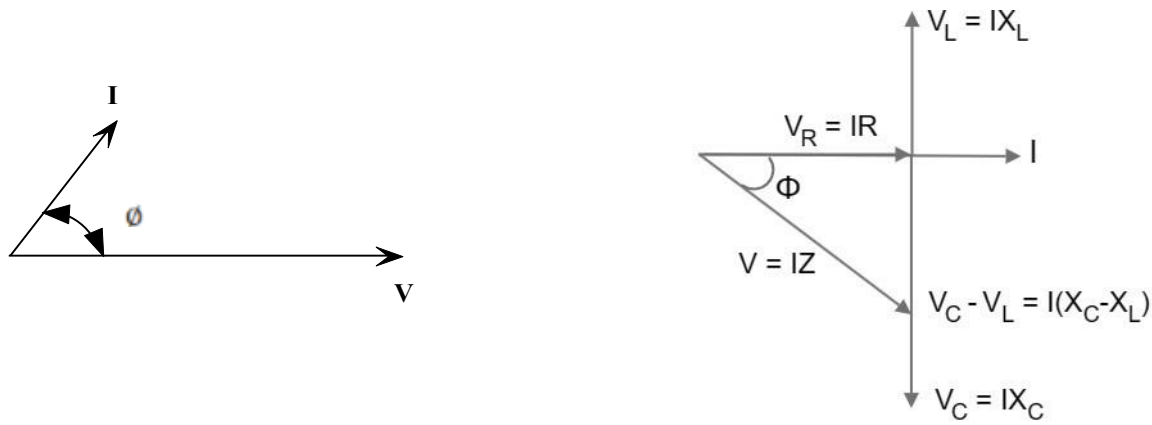


Fig2.14 a) Phasor Diagram for $X_L < X_C$ **b) Impedance triangle**

$$I = \frac{V}{Z}$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

The circuit is similar to an R-C circuit.

If $v = V_m \sin(\omega t)$, then $i = I_m \sin(\omega t + \emptyset)$

Hence it can be proved that the power consumed is given by $P = VI\cos\emptyset$.

Case 3:When $X_L = X_C$

When the inductive reactance is equal to the capacitive reactance, the vector diagram of the circuit as shown in Figure. 2.15.

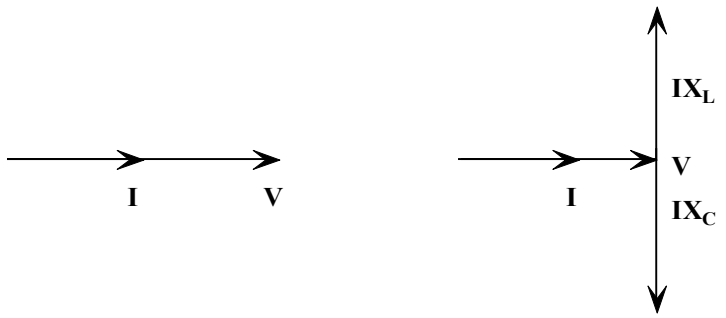


Fig 2.15a) Phasor Diagram for $X_L<X_C$ b) Impedance triangle

V_L and V_C cancel out each other. The current is in phase with the voltage and circuit behaves as a pure resistive circuit. Hence Z = R. The current is in phase with voltage i.e: $\emptyset = 0$ If v= $V_m\sin(\omega t)$, Then ,i=$I_m\sin(\omega t)$

Hence it can be proved that the power consumed is given by P=VI, which is proved in the case of a pure resistance circuit.

Exercise 4: A resistance of 10 Ω is connected in series with a pure inductance of 55mH andthecircuitisconnectedtoa230V,50Hzsinusoidalsupply.Calculatethea) circuit current b) Phase angle

Given data:R=10ΩL=55mHV=230V,f=50Hz

a) Current I= V/Z

X_L= 2πfL=2x π x 50 x 55x10^{-3}= 17.27 Ω

Impedance Z= $\sqrt{R^2 + X_L^2}$

Substituting for X_L, Z= $\sqrt{10^2 + 17.27^2}$ = 19.95 Ω.

∴ Current I =230/19.95=11.52A

b) Phase angle$\emptyset=\tan^{-1}(\frac{X_L}{R})=\tan^{-1}(\frac{17.27}{10})=60$lag(RL circuit)

Exercise 5: A capacitor of 16μ F takes a current of 2 A when the alternating voltage applied across it is 230V.Calculate:(a)the frequency of the applied voltage(b)the resistance to be connected in series with the capacitor to reduce the current in the circuit to 1A at the same frequency(c)the phase angle of the resulting circuit.

a) $X_C = \frac{V}{I} = \frac{230}{2} = 115\Omega$

$X_C = \frac{1}{2\pi fC}$

$\therefore f = \frac{1}{2\pi C X_C} = \frac{1}{2\pi \text{ X } 16 \text{ X } 10^{-6} \text{ X } 115} = 86.5$ Hz.

b) When a resistance is connected in series with the capacitor,

c) $Z=\frac{V}{I}=\frac{230}{1}=230\Omega$

$Z=\sqrt{R^2+X_C^2}=\sqrt{R^2+115^2}=230\ \Omega$

Hence R=199.18Ω

c)$\emptyset=\cos^{-1}\frac{R}{Z}=\cos^{-1}\frac{199.18}{230}=30^\circ$lead (RC circuit)

Exercise 6:A 230V, 50Hz AC supply is applied to a coil of 0.12 H inductance and 5 Ωresistance, the connection in series with a 12μF capacitance. Calculate the impedance, current, phase angle between voltage and current, power factor and the power consumed.

Inductive reactance, $X_L=2\pi fL=2\pi\ X\ 50\ X\ 0.12=37.69\ \Omega$

Capacitive reactance, $X_C=\frac{1}{2\pi fC}=\frac{1}{2\pi\ X\ 50\ X\ 12X\ 10^{-6}}=265.25\ \Omega$

$Z=\sqrt{R^2+(X_C-X_L)^2}=227.6\ \Omega$

I = V/Z = 230/227.6 = 1.01A

To find the phase angle of the resulting circuit:

tan $\emptyset=(X_C-X_L)/R=227.56/5=45.51$

Therefore $\emptyset=88.74^0$ lead

cos $\emptyset$= R/Z = 5 / 227.6 = 0.0219 lead

P = VI cos $\emptyset$ = 230 X 1.01 X 0.0219 = 5.08 W

Exercise 7: A circuit having resistance of 6 Ω an inductance of 0.075 Henry and a capacitance of 50μF in series is connected across a 100V, 50Hz supply. Calculate a) the impedance b) the current c) the voltage across R,L and C d) the phase difference between the current and the supply voltage.

Data given: R=6 Ω ,L=0.075H, C=50μF, V=100V, f=50Hz

$X_L=2\pi fL=23.56\ \Omega$

$XC=1/2\pi fC=63.66\ \Omega$

$Z=\sqrt{R^2+(X_C-X_L)^2}=40.54\ \Omega$

Current I=V/Z= 100 / 40.54 = 2.466 A

Voltage across resistance VR= IR= 2.466 X 6= 14.76 V

Voltage across inductance VL= IX_L= 2.466 X 23.56= 58.09 V

Voltage across capacitance VC = IX_C = 2.466 X 63.66= 156.98 V

Phase difference $\emptyset=cos^{-1}\left(\frac{R}{Z}\right)=cos^{-1}\left(\frac{6}{40.54}\right)=81.48^\circ\ lead$